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La derivada de un producto finito de Blaschke no se anula en la circunferencia unidad

Lema 1. *Sea B un producto finito de Blaschke, entonces* $\forall \zeta \in \mathbb{T} : |B'(\zeta)| \neq 0$.

***Demostración*:** Sea $\zeta \in \mathbb{T}$, entonces por el `lem-derivada-logaritmica-prod-finito-blaschke/`**??**, tenemos que

$$\zeta \frac{B'(\zeta)}{B(\zeta)} = \sum_{k=1}^{n} \frac{|z_k|^2 - 1}{\overline{\zeta}(1 - \overline{z_k}\zeta)(z_k - \zeta)} = \sum_{k=1}^{n} \frac{|z_k|^2 - 1}{\overline{(\zeta - z_k)}(\zeta - z_k)} = \sum_{k=1}^{n} \frac{|z_k|^2 - 1}{|\zeta - z_k|^2} < 0.$$

Como $|B(\zeta)| = 1$ y $|\zeta| = 1$, concluimos que $|B'(\zeta)| = \sum_{k=1}^{n} \frac{1 - |z_k|^2}{|\zeta - z_k|^2} > 0.$ ■

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